

DESIGNING SYSTEMS AND SERVICES FOR REAL WORLD IMPACT

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Royal College of Art

SERVICE DESIGN STAGES

1. **Discover:** gather insights
2. **Define:** frame the right problem
3. **Develop:** ideation with the key stakeholders...
“how might we?” and prototype and test
4. **Deliver:** design and deploy from minimum viable pilot to minimum viable product

Trigger,
competitive,
vision, intent

Clear opportunity,
brief, battle cry

Proven concept
ready for
implementation

Research

- People
- Customer Journeys
- Data
- Safari
- Trends
- Competitors
- Systems
- Operations

Analysis

- Patterns
- Context
- Jobs To Be Done
- Personas
- Pain Points
- Opportunities

Creativity

- Concepts
- Business Models
- Experiments
- Prototypes
- Mock-Ups
- User testing
- Feedback

Specifying

- Impact Metrics
- MVX
- MVP
- Road Maps
- System Change
- Blueprints
- Roadmap

Discovery

Analysis

Brief

Create

Develop

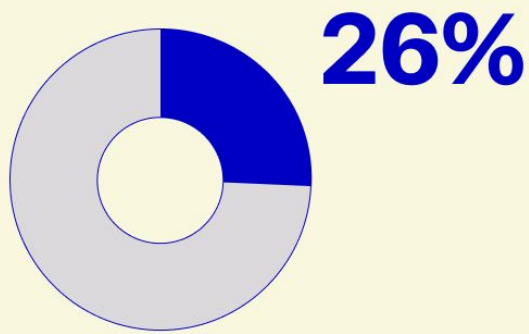
1. DISCOVERY

CURIOSITY ON STEROIDS
IMMERSE YOURSELF IN READING BUT
ALSO THE REAL WORLD

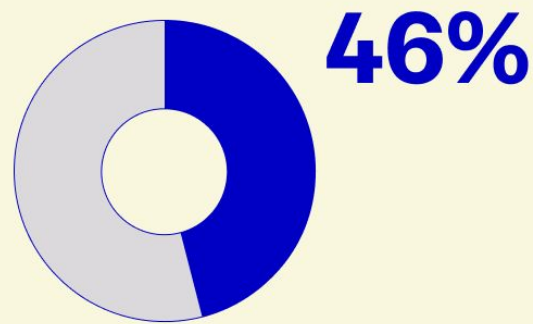
EMPATHY – HUMILITY - INSIGHT

START BY AGREEING THE SCOPE

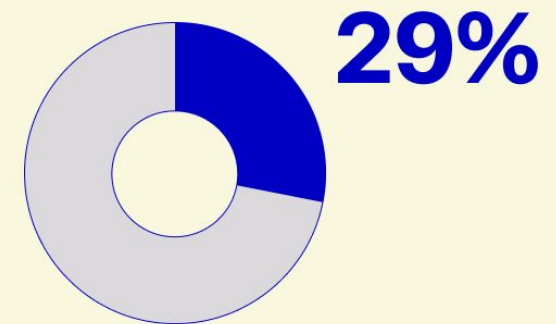




2 billion people
lack safely managed
drinking water



3.6 billion people
lack safely managed
sanitation



3.6 billion people
lack basic hygiene

Source: United Nations Sustainable Development Goals

Week 01 Introduction to Empathetic Engineering

07

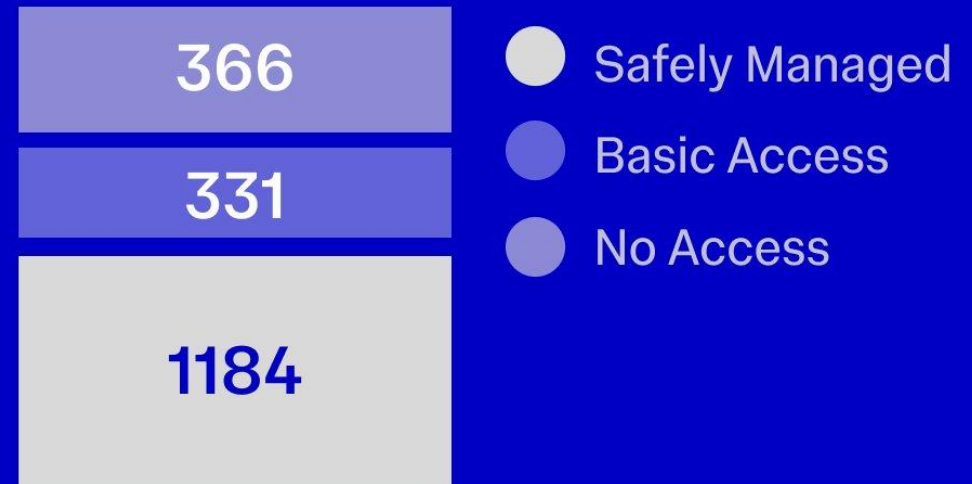
IMPERIAL

Women, Children & those in Emerging Economies are the hardest hit

Primary Responsibility for water collection



Children's access to basic sanitation in millions '19



Full Generic Map



shift^o Obesity System Influence Diagram

Full Map

Clusters

Core Loop

Individual Psychology

Social Psychology

Individual Activity

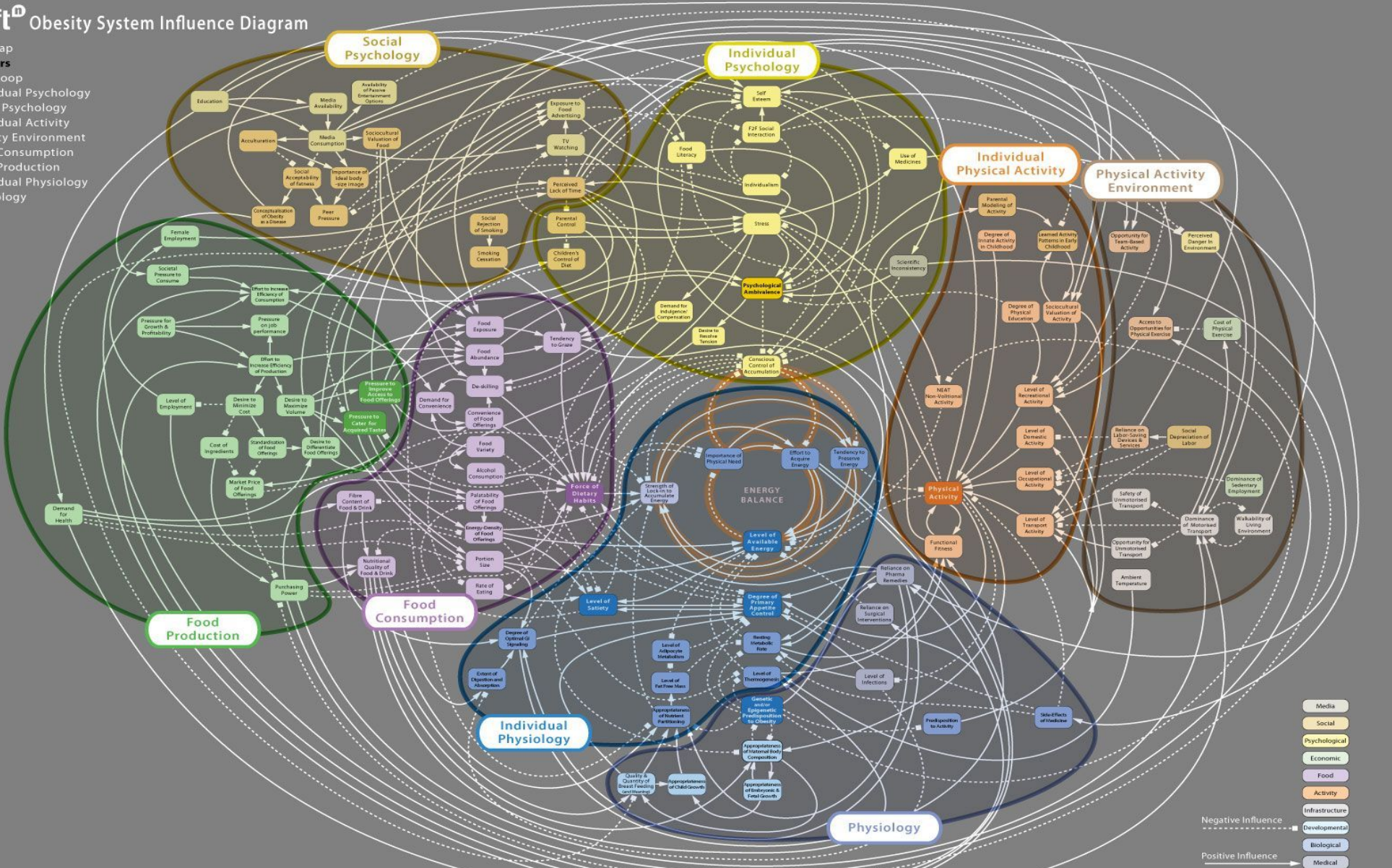
Activity Environment

Food Consumption

Food Production

Individual Physiology

Physiology



MARKET TREND ANALYSIS

Macro-Economic
and Political changes



Observation

Follow people around



Talk and Listen

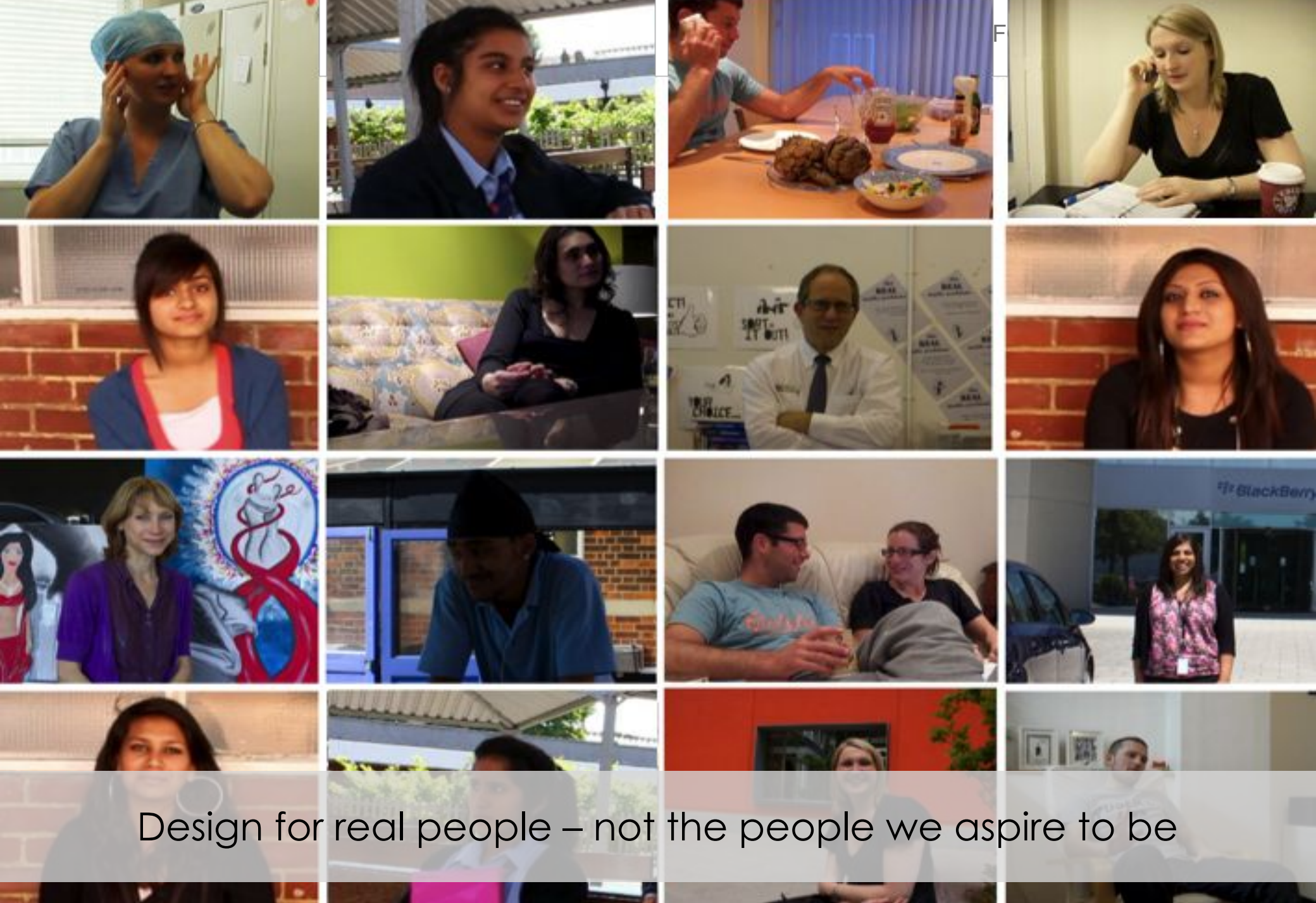


Recreate walk-throughs



PERSONAS

BECAUSE WE CAN'T ALWAYS HAVE
OUR USERS WITH US WHEN WE
DESIGN



Design for real people – not the people we aspire to be

Inclusive





December 3, 1948
£1m+ income
2nd marriage
2 children
Loves dogs



|
December 3, 1948
£1m+ income
2nd marriage
2 children
Loves dogs

Personas go beyond socio-demographics

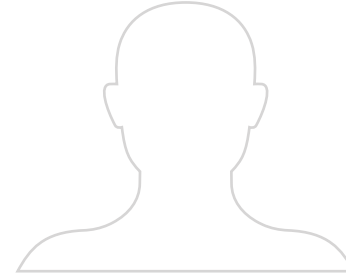
Source: <https://www.linkedin.com/pulse/what-do-prince->



1. Personas Template

Name

Drawing



Job Role, Experience level

Personal Objectives/ Motivations and Ambitions

Frustrations

Desires

Characteristics

Quote

“

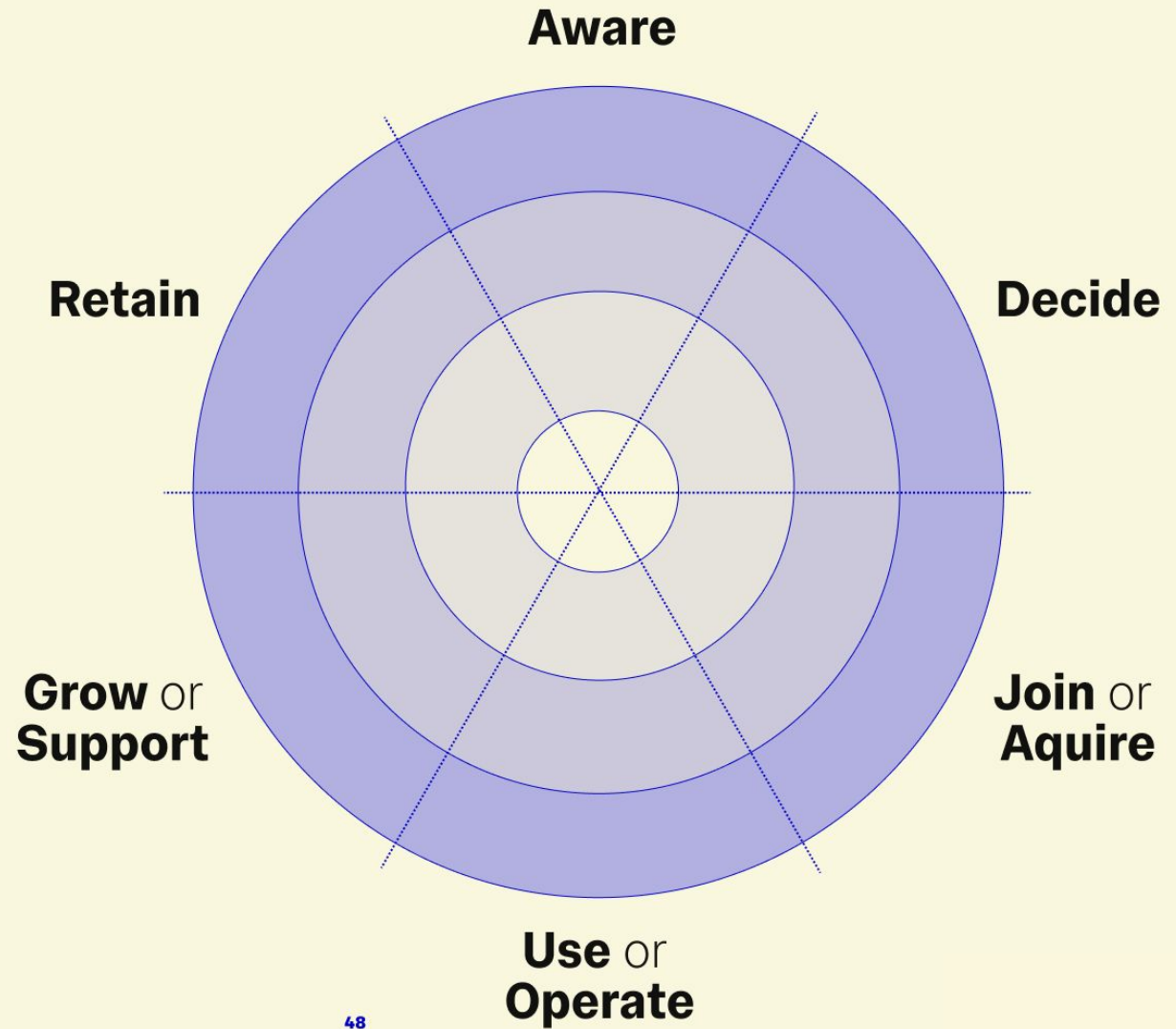
”

If I had a Magic Wand

MAPPING STAKEHOLDERS

Stakeholders

Understand key stakeholders at each stage in the lifecycle



2. DEFINE

MAP THE CUSTOMER OR USERS JOURNEY OVER TIME

HOW DO THEY DO IT TODAY AND WHAT ARE THE
PAINS? WHAT COULD BE THE GAINS?

Journeys & Touchpoints



- 01 Metaphor that allows us to **chart the user experience**, as opposed to process mapping
- 02 Through this we can identify **overlap/gaps and highs/lows**
- 03 Opens up the possibility to **discuss opportunities with users**
- 04 **Steps by time**, running left to right.
Before, during and after

Positive
Emotions

Negative
Emotions

We can overlay users'
emotions to journeys to
establish highs and lows
and where to focus



A series of post its can convey even the most complex of customer journeys

Lock up at home

Get taxi to airport

Check which terminal

Go through the terminal door

29

Lock up
at home

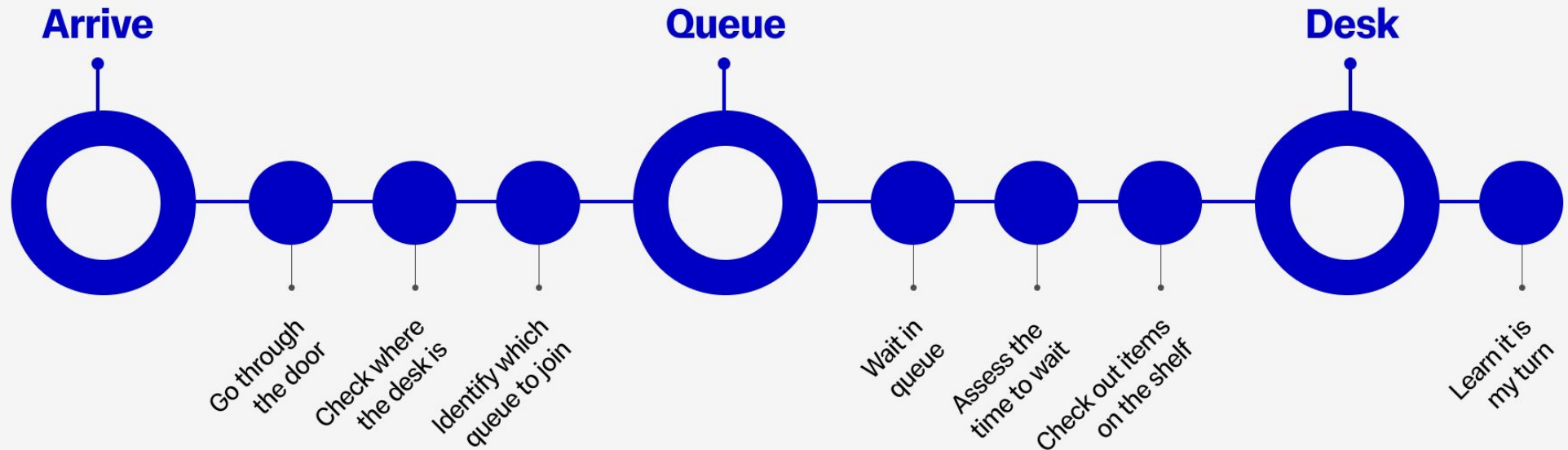
Get taxi
to airport

check which terminal

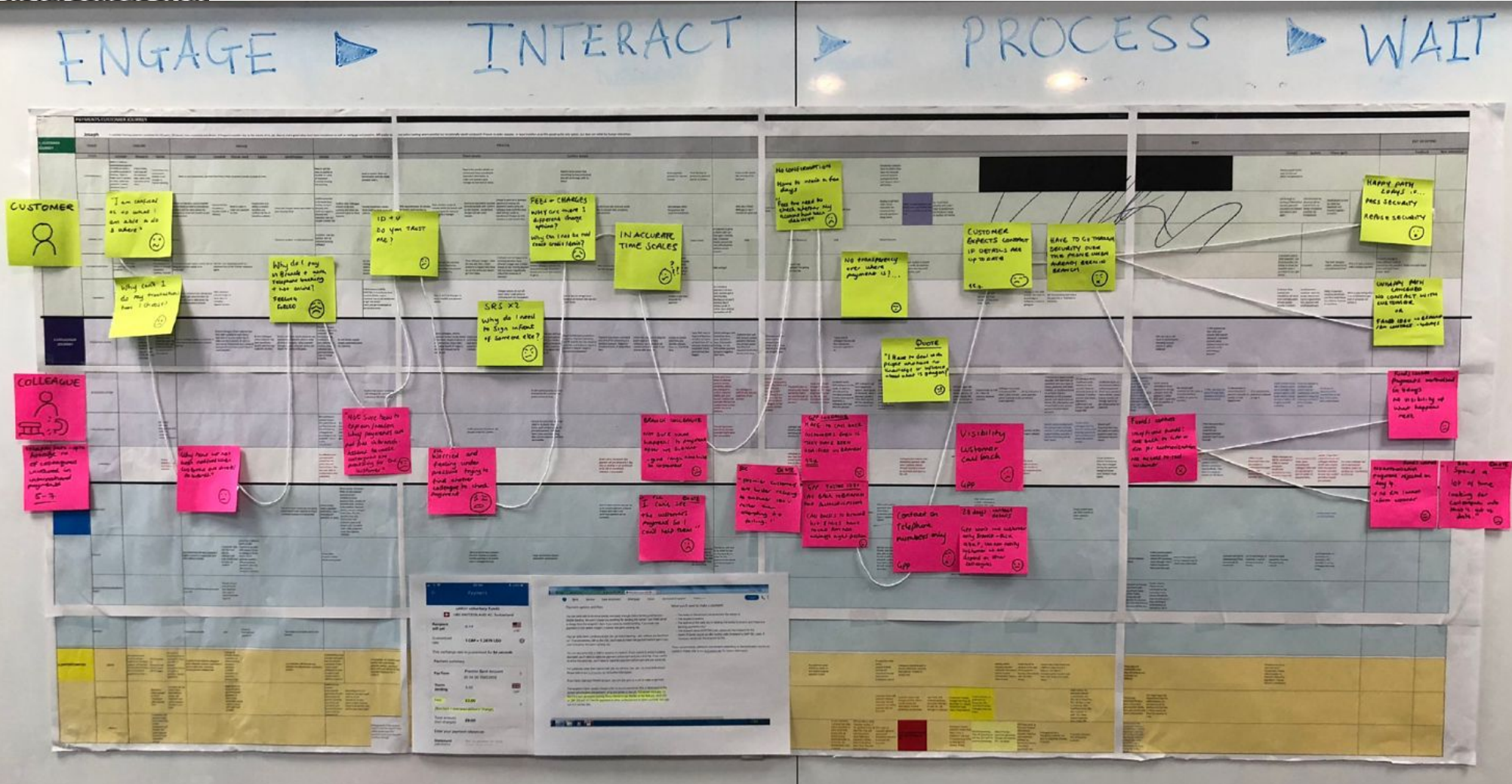
Go through
the terminal
door

Customer Journey Mapping

Analyse every step of the journey and establish what the customer experiences as a series of **major and minor touchpoints**





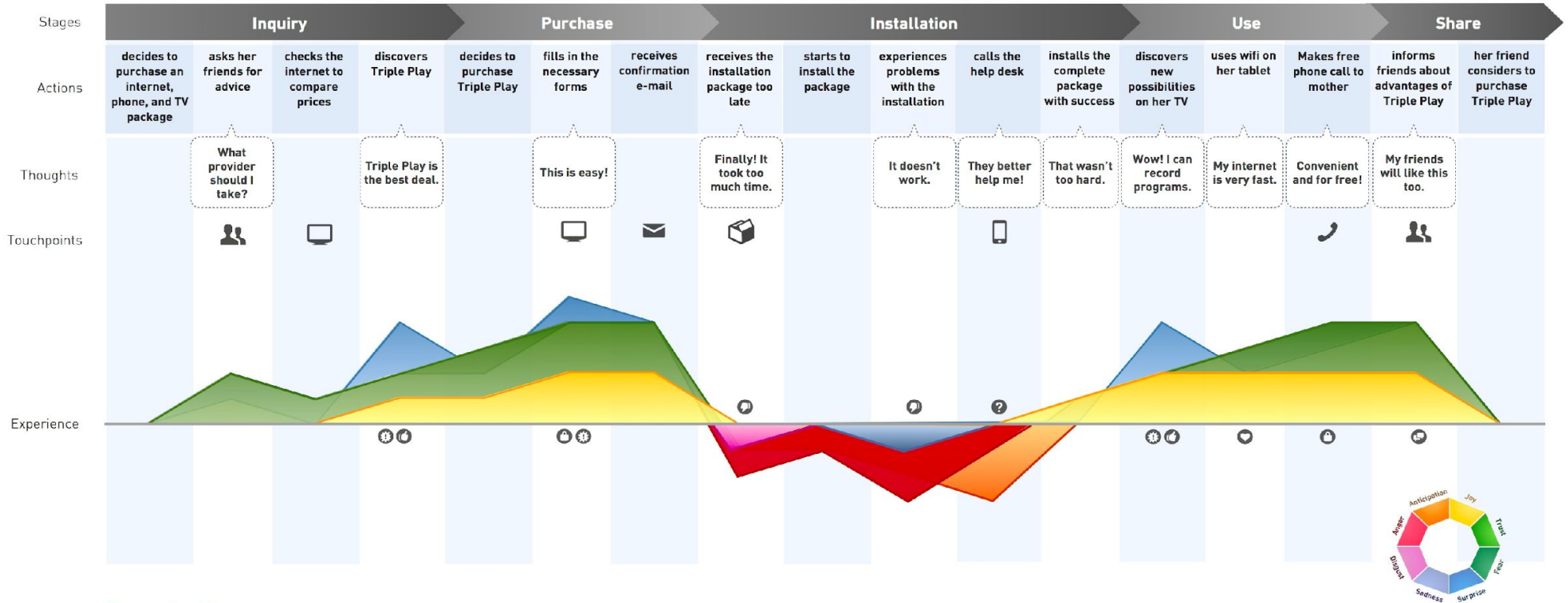




Eva is moving to Amsterdam. She needs phone, internet and TV services. She will base her choice based on price, availability and previous experience with service providers.

CUSTOMER EXPERIENCE JOURNEY MAP

Customer Experience Map



Opportunities

Attract and Convert

Emphasize the benefits of the Triple Play package.

Stage: Inquiry

Use different channels to advertise the Triple Play package.

Stage: Inquiry

Provide online comparing possibilities with other services.

Stage: Inquiry

Attract attention on social media such as Facebook.

Stage: Inquiry

Enable discussions on social media about services.

Stage: Inquiry

Support

Reward new customers during the waiting process.

Stage: Purchase

Provide clear information about delivery times repeatedly.

Stage: Purchase

Provide professional, attentive, and friendly support for free.

Stage: Installation

Provide clear and easy to find do-it-yourself manuals.

Stage: Installation

Provide easy to reach support through different channels.

Stage: Installation

Share

Offer unique features with the Triple Play package.

Stage: Use

Reward customers who use Triple Play intensively.

Stage: Use

Enable customers to easily recommend Triple Play online.

Stage: Share

Reward customers who recommended Triple Play.

Stage: Share



THE JOURNEY BEGINS

YOUNG PEOPLE WITH CARE EXPERIENCE

JOURNEY THROUGH THE CRIMINAL JUSTICE SYSTEMS

AMBITIOUS FOR THE FUTURE TOGETHER

Adverse Childhood Experiences

Peer relationships experiment with crime

SCHOOL

School Exclusion

CHILDRENS HOMES

HIGH DEPENDENCY LOW EMPATHY SYSTEM

Childrens Home/Foster parents – criminalisation of behaviour incidents

Police engagement which escalates

SILOED RESPONSIBILITIES

FRAGMENTED VIEW

TURNING 18:

- Expected to live independently
- Staff to child ratio disappears
- Dropped into the adult system

THE FIRST CLIFF EDGE

THE ADULT SYSTEM

BUT THERE IS A BETTER WAY!

RESTORATIVE PRACTICE

Short and long term goals

THE ASPIRATION ENGINE

Aim high!

TRAUMA ATTACHMENT

Professional training on care experience linked to trauma and attachment

THE FALL

Leaving custody with no relationship support

Prison

THE SECOND CLIFF EDGE

No place to live

Drugs

Suicide risk

COLLABORATION LEVELS

National Regional Local

Ministers

COLLABORATION INDICATORS

- Universal family
- Empathy
- Young person centred training

Measuring the interaction of services up-lifting the experience of young people with care experience

Resolve

SPECTRA⁺

Seriberia

CUSTOMER JOURNEY TEMPLATE

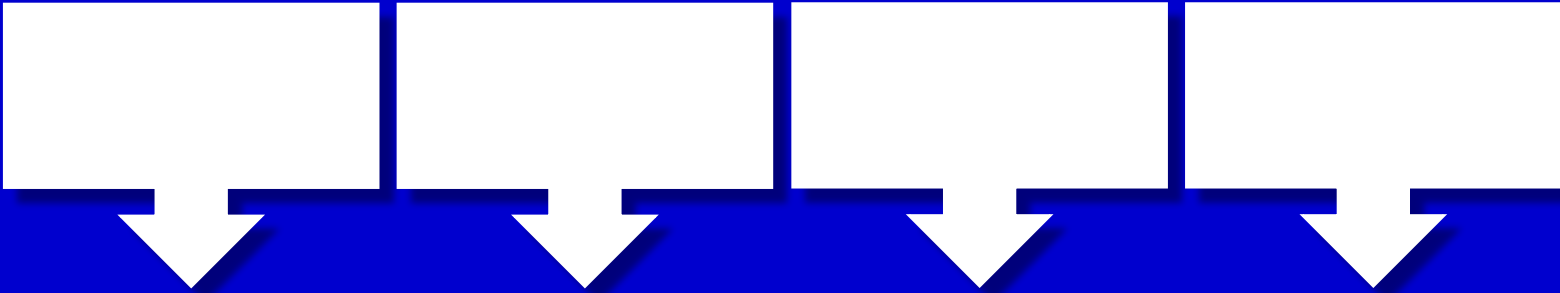
		AWARE	DECIDE	JOIN	FIRST USE	REPEAT USE	SUPPORT	RETAIN
KEY STEPS								
USER EXPERIENCE	High							
	Average							
	Low							
CHALLENGES								
OPPORTUNITIES								

MISSION STATEMENT

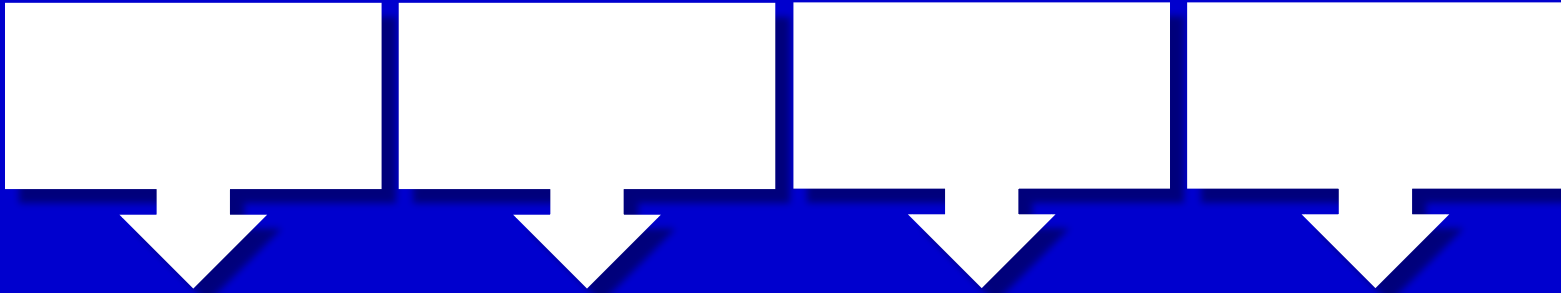
SET YOUR NORTH STAR

MISSION STATEMENT

CHALLENGES



OPPORTUNITIES



MISSION STATEMENT

Use your findings from the Heat Map to outline a problem statement to ensure you are solving the right problem

Translate the HMWs into a problem statement that describes for whom, the opportunity, and so that...

3. DESIGN: IDEATION

HOW TO COME UP WITH LOTS OF IDEAS FAST

.... AND NOT JUST YOUR OWN

A group of people are seen from behind, standing in a meeting room. They are looking at a wall covered with numerous sticky notes, suggesting a collaborative brainstorming or ideation session. The image has a blue tint.

Ideation and
Concept Creation.
**Creativity is an act
of Collaboration →**

Have many ideas, and Then have some more →



Rolling Bridge, Paddington by Thomas Heatherwick

Source: <https://www.dezeen.com/2015/12/18/dezeen-a-z-advent-calendar-rolling-bridge-thomas-heatherwick-london/>



01. Brainstorming

02. **Crazy 8's**

- Silent and Solo
- Fold Paper 3 times
- Based on your How Might We
- 1 minute, draw an idea
- Then draw another
- 8 minutes, 8 ideas
- You can do it...

Ideate using **Crazy 8's**



Week 05 Ideate

09

03. Parallel Worlds



AWARD WINNING REDESIGN OF LONDON AMBULANCE FROM
FORMULA 1 PIT STOP STUDY

04. Future Envisioning



FINANCIAL TIMES

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Edward Luce
How Obama lost friends and influence

Mohamed El-Erian
Risks in buying Russian bonds on dips

Cleaning up Did BP's spill really kill the Gulf of Mexico?
Andrew Gimson Religion is the prime minister's backbone
Lex on toy makers The business of child's play

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Highlights • Interview: Bidco's Vimal Shah • Cringe-free personal statements on CVs • Power in the handbag • In Depth: The Fragile Middle



JP Morgan: the future starts here

JP Morgan is setting the pace in the use of technology to deliver super-human advice that's helping its clients generate returns that are outstripping its competitors. The next generation of Fin Tech companies are now chasing JP Morgan as it sets the pace in exploiting the very best in AI and Data Science combined with personal service excellence. And its delivering at scale. JP Morgan has shown itself to be a truly multi-asset solution provider and its global network of innovation centres have helped it to go to the place for its partner eco-system as together they design the future of financial services.

5:48am
Japan posts largest-ever trade deficit
\$134bn gap underscores wrenching structural shift for economy

4:38pm
Brown to join Scotland unionist campaign
Former PM gives his support as polls narrow

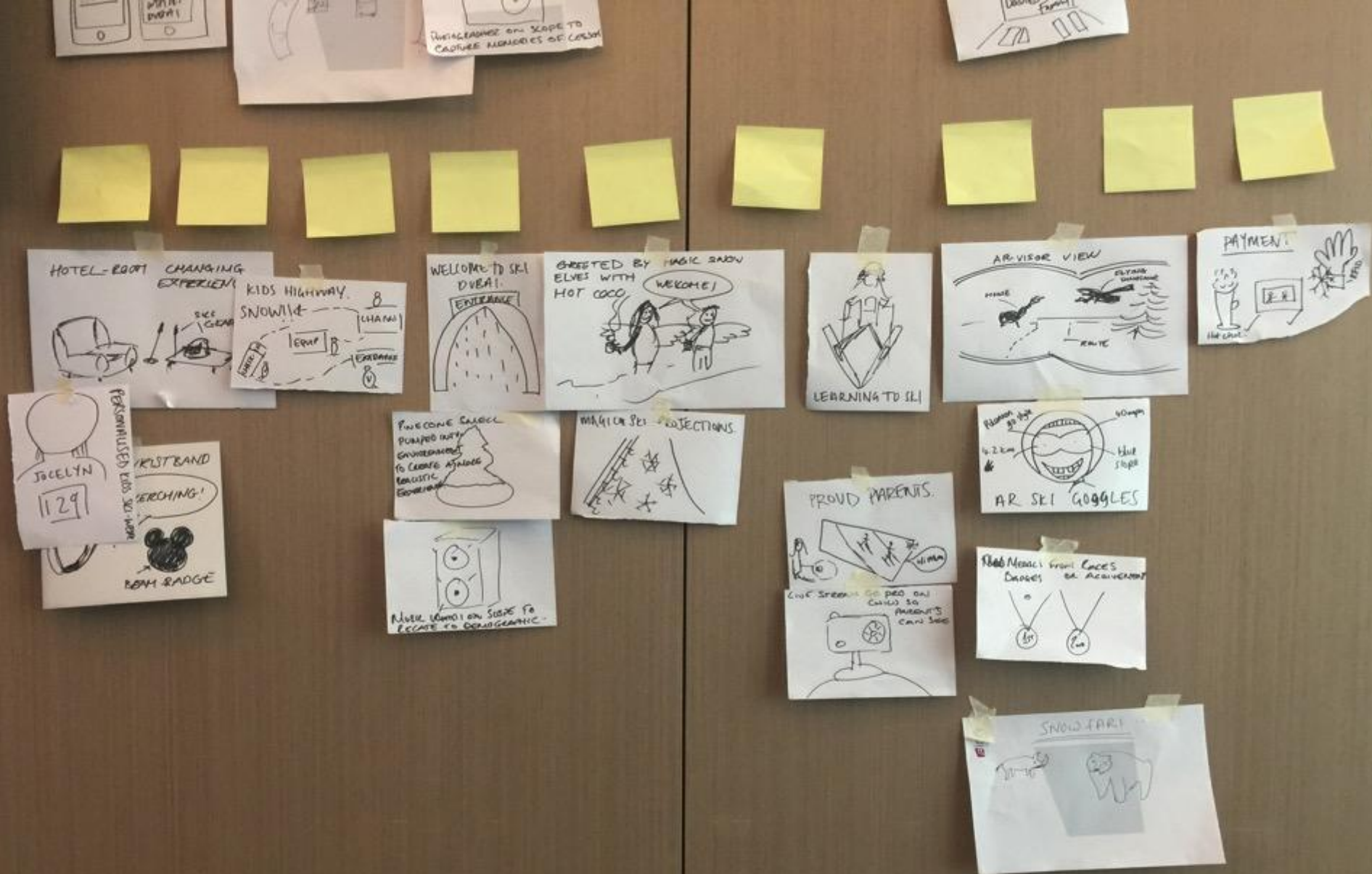
6:45am
Fraud fears grow in Afghan poll count
Winner still unclear and thousands of votes are under scrutiny

9:33am
Japanese ship seized in wartime claims row
Winning bid for reparations

FAST FT		MARKETS DATA	
Indices		Commodities	Currencies Bonds
FTSE 100	6,625.25 +0.62%		
S&P 500	1,864.85 +0.14%		
Eurofirst 300	1,328.71 +0.47%		
Nikkei 225	14,512.38 -0.03%		
Shanghai	2,065.83 -1.52%		
Data delayed by at least 15 minutes		Portfolio Markets Data	



BACK-CASTING FROM THE INSPIRED FUTURE VISION



Applying the ideas to reimagine the customer journey



A close-up photograph of an open sketchbook. The left page features a storyboard layout with several rectangular frames. The bottom-left frame contains a simple line drawing of a smiling face. The right page shows a large, hand-drawn diagram with curved lines and arrows, possibly representing a process or a narrative flow. A black pen is visible in the upper right corner, and a red bookmark is at the bottom of the right page.

Storyboarding



Creating a Storyboard of our Service Experience

AIRBNB



Guest



Sign up



Find listing



Book



Check-in



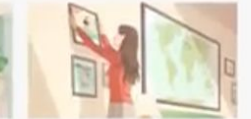
Stay



Check out



Review Host



Get rewarded & develop

Host



Sign up



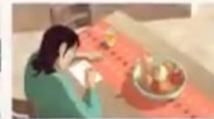
Publish listing



Get booked



Provide check-in



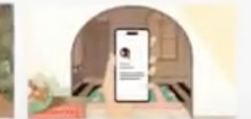
Host



Provide check out



Review guest



Get rewarded & develop

Persona:

User Story / Scenario:

PROTOTYPING AND TESTING

SEE WHAT WORKS – FAIL FAST TO LEARN FASTER

Prototyping a Product or Service

User Experiences

User Journeys, and Stories -
Hearing, Trying, Buying, Using and
Sharing - Emotions

Touchpoints

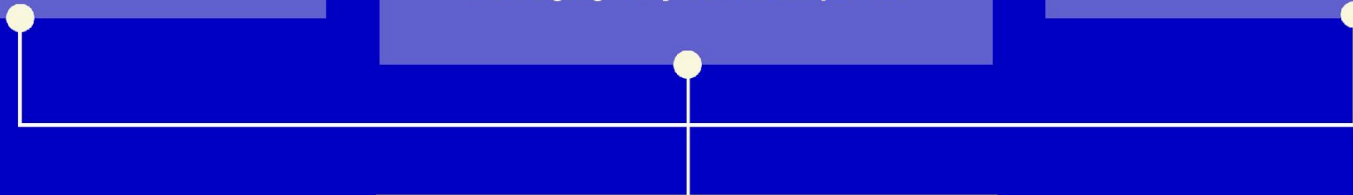
People and their Environments -
Products, Tools and Technologies -
Brand, Communication and
Packaging - Physical Touchpoints

Operations

Process - Workflows - System
Interactions - Information/ Data -
Organisational Interactions

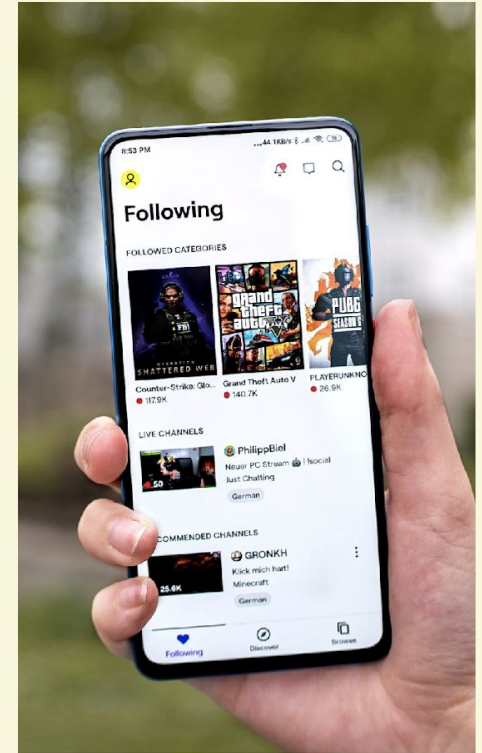
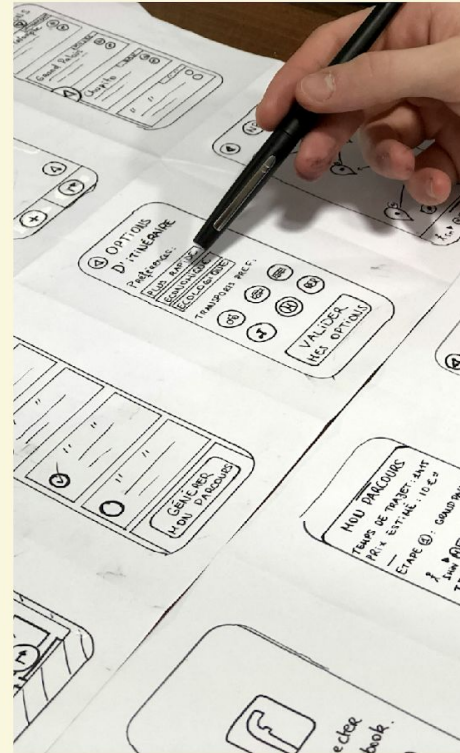
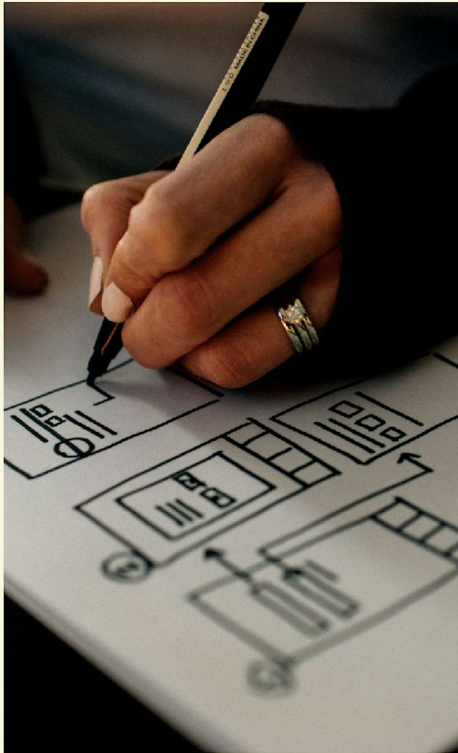
Techniques

Sketches and Storyboard - Digital
Design Tools - Physical Models -
Software Prototyping



Prototyping & Testing

Different Levels

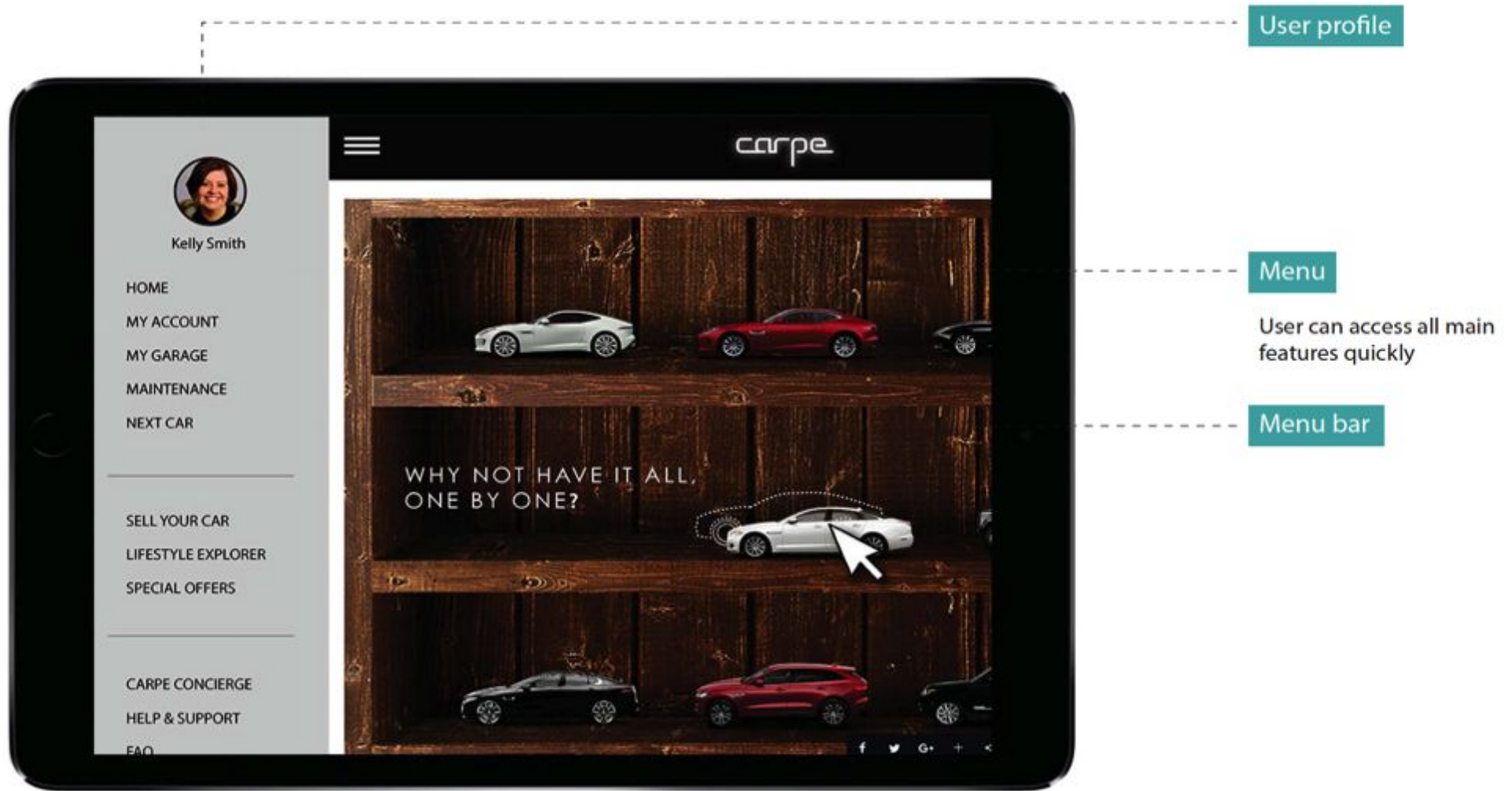


Prototyping & Testing

- > From low-fi to hi-fi
- > Test your understanding of the systems context, the stakeholders
- > Check the validity of your assumptions not just the solutions you've envisioned

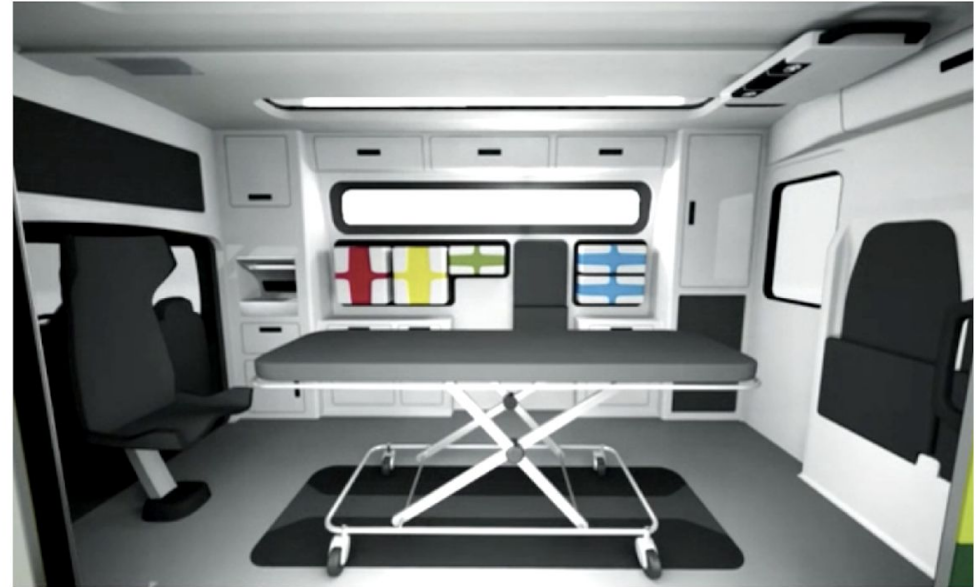


Key screen / home menu



Translate into operational prototypes with high fidelity

Immersive VR Digital Prototype



Source: Helen Hamlyn Centre - Redesigning the Emergency Ambulance

3D Physical Prototype

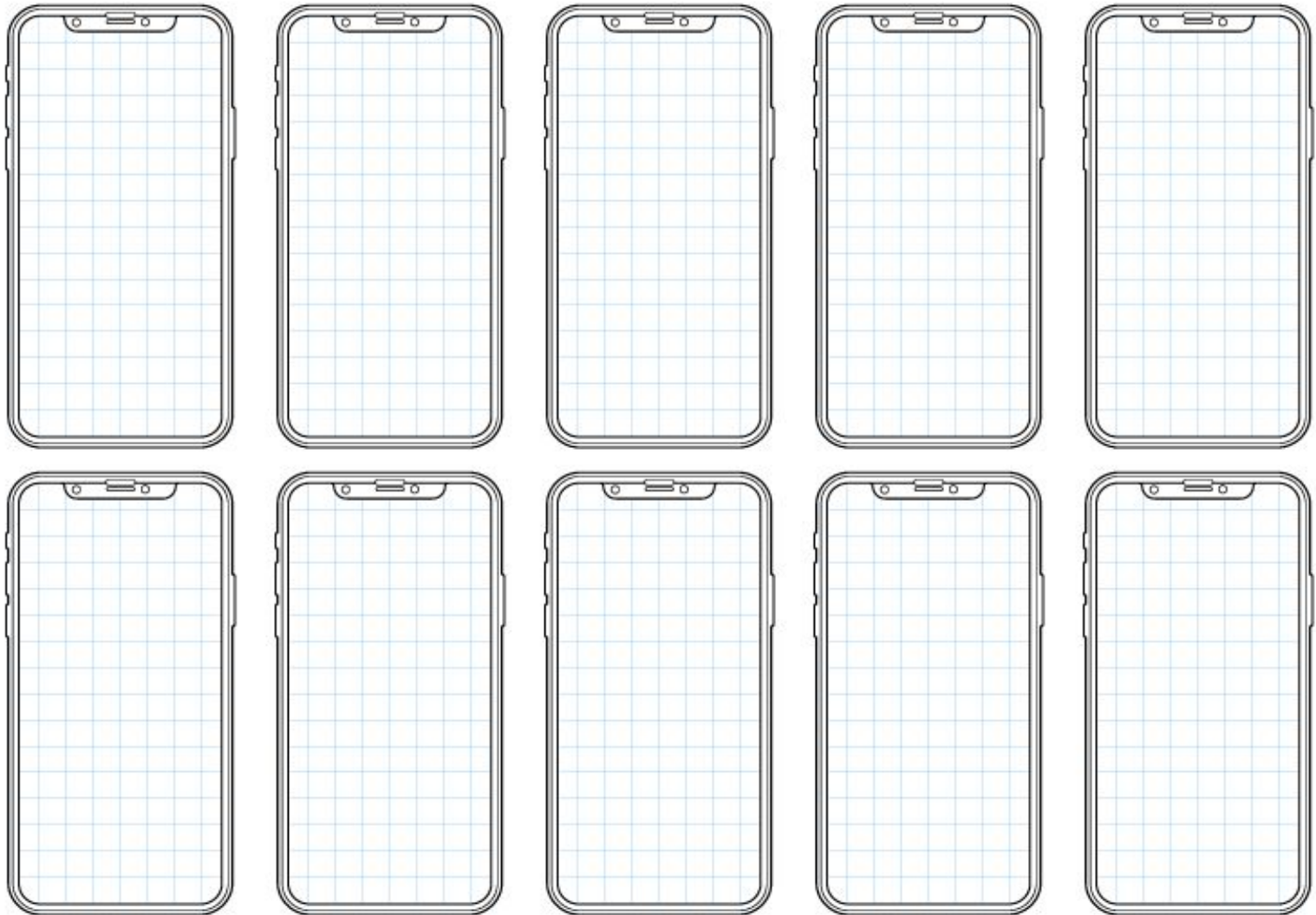


Source: Helen Hamlyn Centre - Redesigning the Emergency Ambulance

Physical Model with Digital Systems



Physical Models to test with users including digital communication and monitoring system, connected with data and video link to hospital



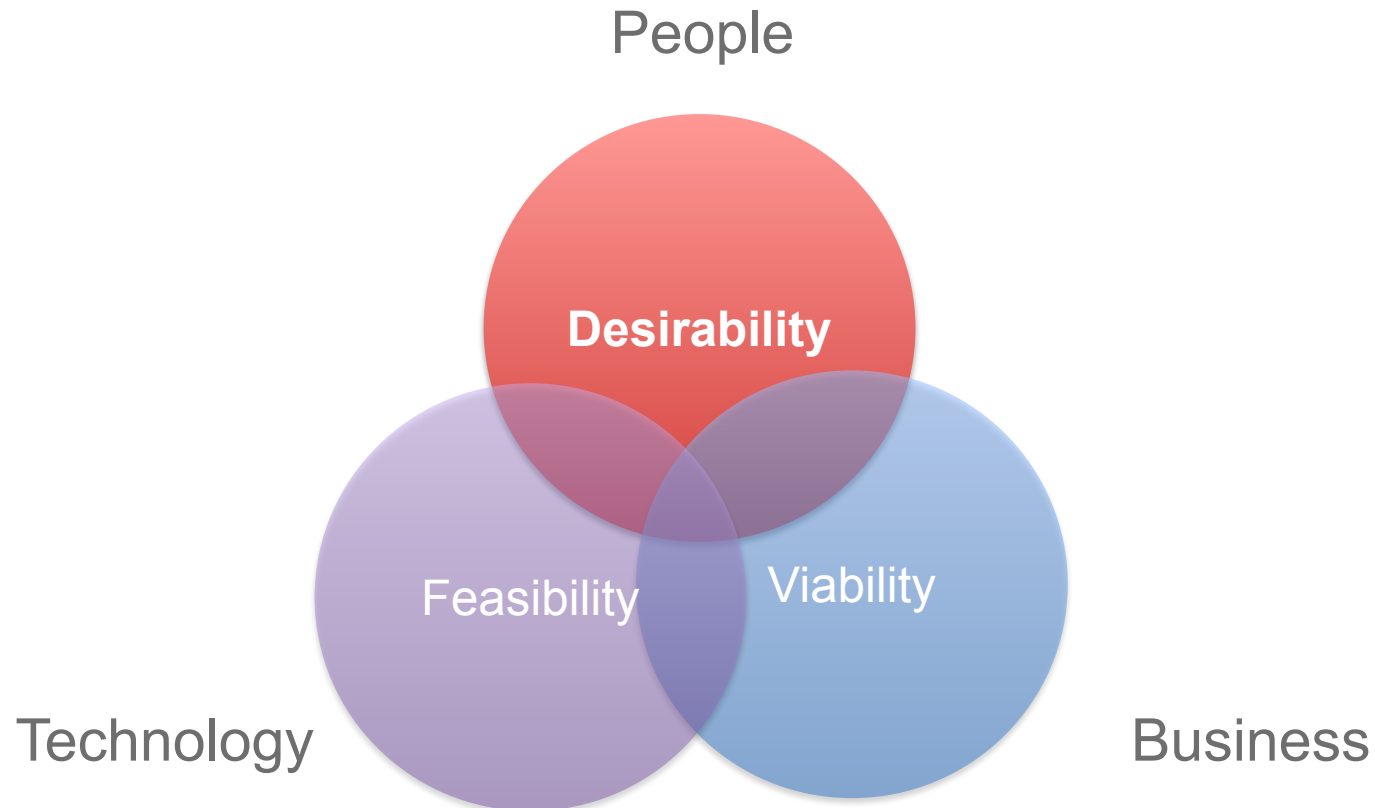
.....and developing some wire frames

4. DELIVERY

DESIRABILITY – FEASIBILITY – VIABILITY

What will be the new business model?

A balanced approach to innovation



Our help(s) who want to by and (unlike

NOW SUMMARISE IT AND TRANSLATE THE CANVAS INTO A VALUE PROPOSITION STATEMENT

Our discover weekly feature of our Streaming service
Products and Services

help(s) Non expert music listeners
Customer Segment

who want to Have fun by getting to know new music

by Giving them bespoke playlists every week
jobs to be done

and Curating music based on their own preferences
verb (e.g., reducing, avoiding)

(unlike Other services, Spotify is natively connected with your own social networks and devices of any brand
and a customer pain

) Expanding their music knowledge and taste
verb (e.g., increasing, enabling)

and a customer gain

Simplified Business Model

Supply Chain Partners
and
Your Core Capabilities

Value
Proposition

Markets
and How you
Serve Them

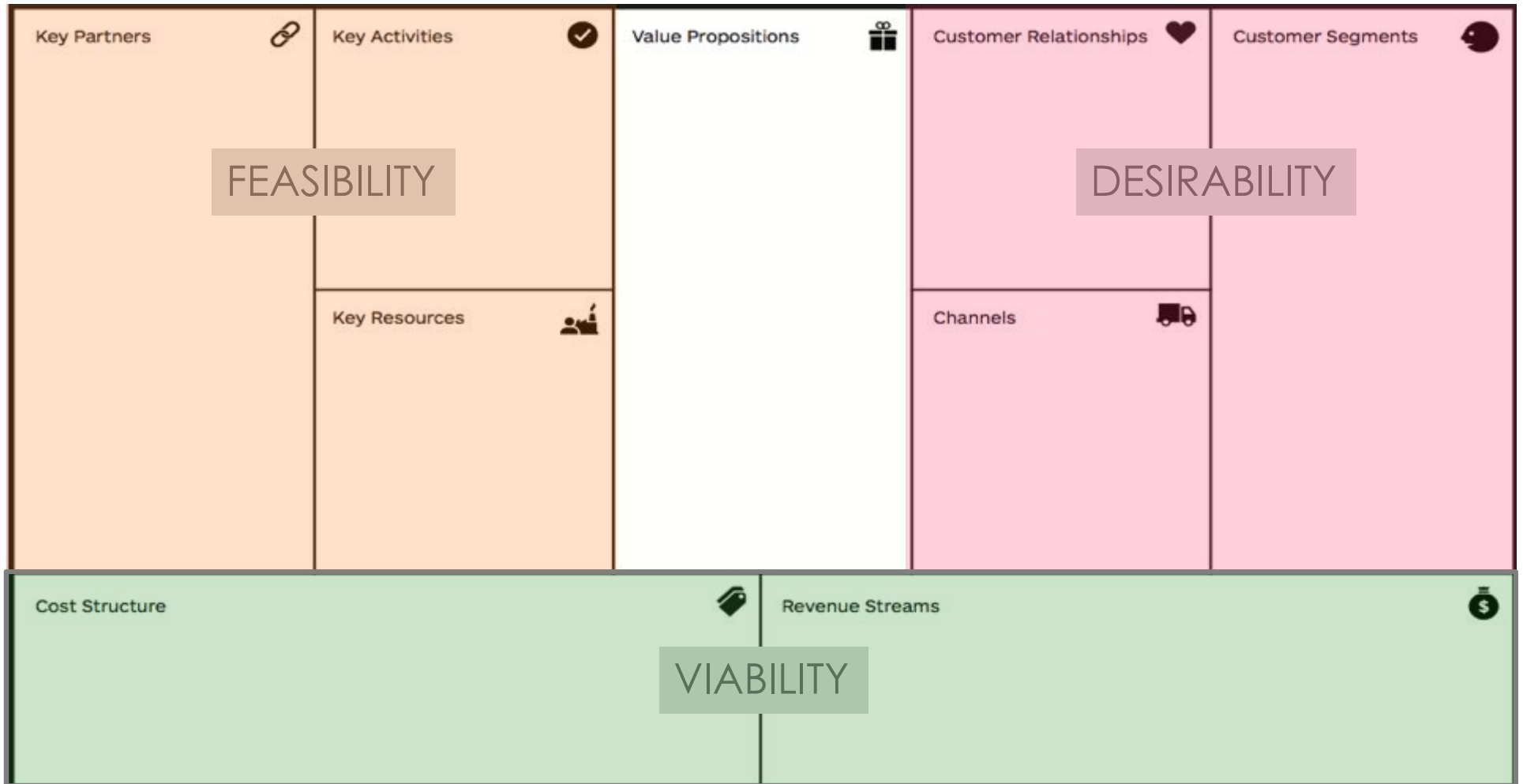
Creating
Value

Managing
your Cost

Generating
Income

Capturing
Value

ASSESS AND SELECT DIFFERENT CONCEPTS USING THE BUSINESS MODEL CANVAS



Principles of Systems and Service Design

- **Empathy:** Understand your users, their values, their real not imagined needs, and their lives
- **Orientation:** Innovate from the outside in, rather than from the inside out
- **Systems Thinking:** Systemic response to systemic issues
- **Alignment:** Design the customer experience then engineer the products, processes and systems
- **Collaboration:** Value is co-created and that demands co-design
- **Experimentation:** prototyping to test understanding of the problem not just the solution